INCH-POUND

MIL-I-25859C(USAF)
30 January 1995
SUPERSEDING
MIL-I-25859B(USAF)
24 September 1971

MILITARY SPECIFICATION

INDICATOR, THRUST AEU-2/A, SYNCHRO STYLE, PRESSURE RATIO, 1.2-3.4 UNITS, INTEGRALLY LIGHTED

This specification is approved for use by the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of integrally lighted, synchro style, pressure ratio thrust indicator designated AEU-2/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-1672	Boxes, Shipping, Reusable With Cushioning

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-I-7057	Indicator, Synchro, Aircraft, General
MIL-I-25467	Lighting, Integral, Red, Aircraft, Instrument, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/ TICLA, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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MIL-T-25485	Transmitter, Thrust, Single Synchro Style
MIL-B-27497	Bearing, Jewel, Sapphire Or Ruby, Synthetic

STANDARDS

FEDERAL

FED-STD-101	Preservation, Packaging And Packing Materials: Test Procedures
FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements
MS33585	Pointers, Dial, Standard Design Of Aircraft Instrument
MS33678	Connector, Receptacle, Electric Integral Mounting

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 General specification. The requirements of MIL-I-7057 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.2.1 Exceptions to the requirements of MIL-I-7057. The high altitude-low temperature test of MIL-I-7057 shall not be required by this specification.

3.3 Design and construction. The indicator shall be designed to operate on the synchro system as specified in MIL-I-7057. Unless otherwise specified, the indicator shall be designed to indicate remotely the pressure ratio of a turbo-jet engine when used in conjunction with a pressure ratio transmitter in accordance with MIL-T-25485.

3.3.1 Case. The case shall conform to figure 1 and shall be hermetically sealed. The length of the case, excluding the connector, shall be 4.75 inches maximum.

3.3.1.1 Filling medium. The case shall be filled with 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The filling medium shall be of at least 90 percent purity, free of dust particles, and shall contain not more than 0.006 milligrams of water vapor per liter (dew point -65° C) at the filling pressure. The absolute pressure of the medium shall be approximately one atmosphere.

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3.3.2 Dial. The dial shall conform to figure 2. The pressure ratio versus angular degrees shall be linear with $\pm 0^{o}$ 15' of arc. The dial diameter shall be the maximum possible consistent with good design practice.

3.3.2.1 Dial to cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.200 inch.

3.3.2.2 Dial markings. The following markings shall be finished in lusterless white, color 37875 of FED-STD-595. The dimensions of the markings shall be as follows:

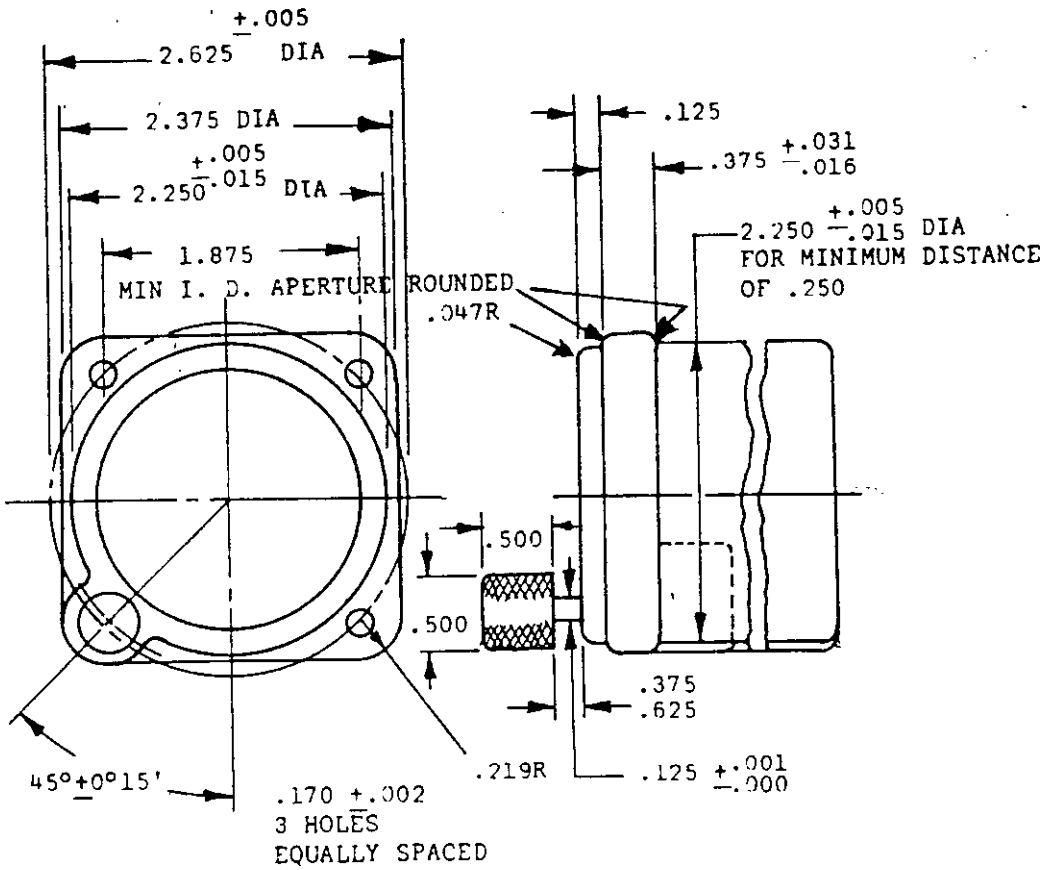
MARKING	HEIGHT OR LENGTH INCH ± 0.016	WIDTH OF LINE OR GRADUATION INCH ± 0.005
Numerals 1.5, 2.0, 2.5, and 3.0	0.156	0.020
Numerals 1.2 and 3.4	0.125	0.016
Counter numerals	0.125	0.016
Major graduations	0.188	0.020
Minor graduations	0.156	0.016
Lettering ENG., PRESS, RATIO	0.094	0.016
Lettering CRUISE	0.125	0.016
Lettering TAKE-OFF	0.125	0.016
Electrical zero reference mark	0.156	0.016

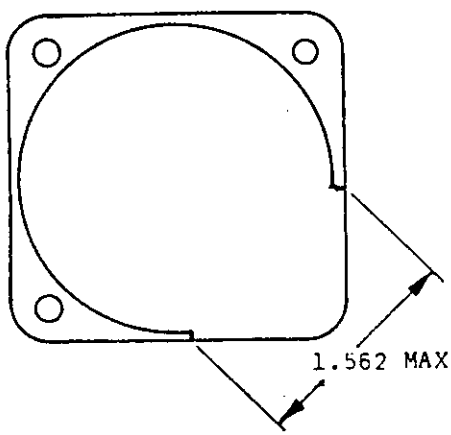
3.3.3 Counters and indices.

3.3.3.1 Takeoff index and counter. The dial shall contain a moveable index and a drum-type counter for takeoff-pressure-ratio reference indication. The index shall conform to figure 2 and shall be mechanically related to the counter reading in pressure ratio as shown in figure 2. The maximum allowable position error between the counter and the center V of the index shall be ± 0.005 pressure ratio units. The counter shall indicate pressure ratio from 1.20 to 3.40. The index and associated counter shall move smoothly over the complete scale range of 1.20 to 3.40 pressure ratio when a torque of 6 ±3 inch ounces is applied to the control knob. The index and counter shall be moved by the indexing mechanism knob when the knob is pushed and turned. The length of the index shall be 0.156 + 0.000 or - 0.031 inch, depending on the pointer length (see 3.3.4). The counter numerals shall move downward with an increasing pressure ratio as the knob is rotated clockwise. The mechanism employed for driving the index and counter shall be rugged in design. The control knob and drive shaft shall conform to the dimensions specified in figure 1. There shall be no binding, deformation, or malfunction when the indexing mechanism is engaged and rotated over its design range. Suitable ruggedclutch mechanism shall be provided to prevent the index and counter from being damaged or mismatched if rotated above or below the dial range.

3.3.3.2 Cruise index and counter. The dial shall contain a moveable index and a drum-type counter for accurate cruise-pressure-ratio reference indication. The index shall be mechanically related to the counter reading in pressure ratio as shown in figure 2. The maximum allowable position error between the counter and the center line of the index shall be ± 0.005 pressure ratio units. The counter shall indicate pressure ratio from 1.20 to 3.40. The index and associated counter shall move smoothly over the complete scale range of 1.20 to 3.40 pressure when a torque of 6 ± 3 inch ounces is applied to the control knob. The index and counter shall be moved by the indexing mechanism knob when the knob is pulled and turned. The length of the index shall be 0.156 + 0.000 or -0.031 inch, depending on the pointer length (see 3.3.4). The counter numerals shall move downward for increasing pressure ratio as the knob is rotated clockwise. The mechanism employed for driving the index and counter shall be rugged in design. The control knob and drive shaft shall conform to dimensions specified in figure 1. There shall be no binding, deformation, or malfunction when the indexing mechanism is engaged and rotated over its design range. Suitable rugged stops or slip-clutch mechanism shall be provided to keep the index and counter from being damaged or mismatched if rotated above or below the dial range.

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REAR VIEW

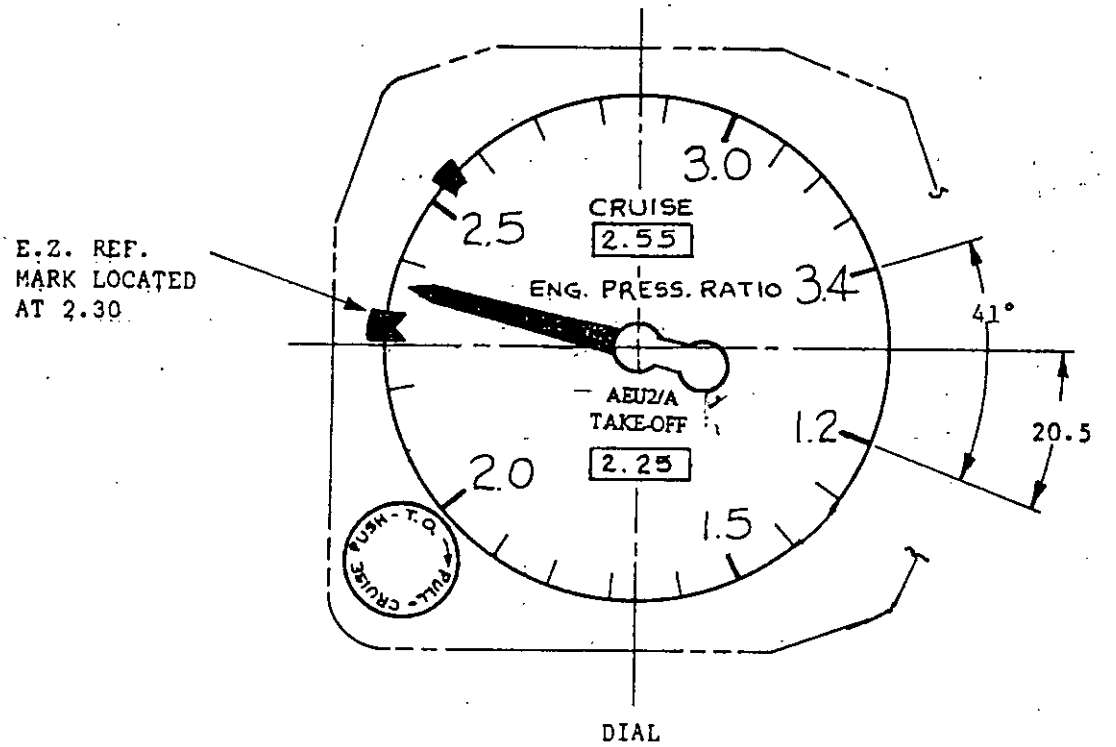
FRONT AND REAR MOUNTING FACES SHALL BE FLAT WITHIN ±.005. ALL SCREWS AND NUTS, EXCEPT MOUNTING NUTS SHALL BE FLUSH WITH MOUNTING FACES

FOR INSTALLATION, CLEARANCE SHALL BE ALLOWED FOR THE MAXIMUM OVER-ALL DIMENSIONS SHOWN ON THE DRAWING.

DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED. TOLERANCES: DECIMALS ± .016

FIGURE 1. Case.

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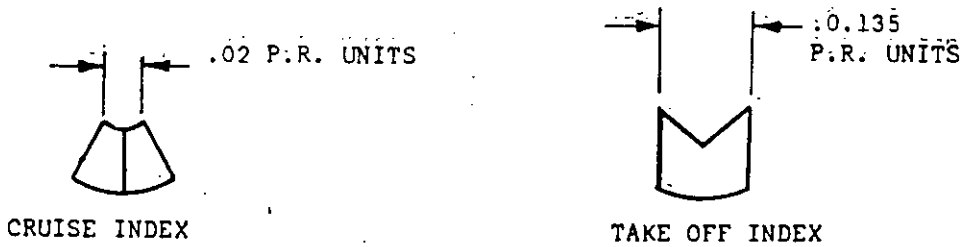


FIGURE 2. Index and dial.

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3.3.4 Pointer. The pointer shall conform to MS33585-3, except that the taper at the pointer shall extend 0.3125 inch and the length of the pointer shall be such that the tip extends to a distance not more than 0.020 inch from the nearest pointer on the indexes. The pointer shall overlap the shortest graduations by a length equal to not more than 0.20 the length of the graduation.

3.3.5 Knob. The knob shall conform to figures 1 and 2, and shall be finished in lusterless black. The lettering shall be finished in lusterless white, color 37875 of FED-STD-595, and shall be as large as possible. The sides of the knob shall be knurled.

3.3.6 Electrical receptacle. The electrical receptacle shall conform to MS33678 R-14S-5P. Power for the lighting system shall be fed through pins A and E.

3.4 Indicator to transmitter wiring. The external wiring shall be in accordance with figure 3.

3.4.1 Weight. The weight of the indicator shall not exceed 1.5 pounds.

3.5 Performance. Performance shall be as specified in MIL-I-7057.

3.6 Integral lighting. The indicator shall be integrally lighted in accordance with MIL-I-25467.

3.7 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.8 Bearings synthetic. When sapphire or ruby jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.9 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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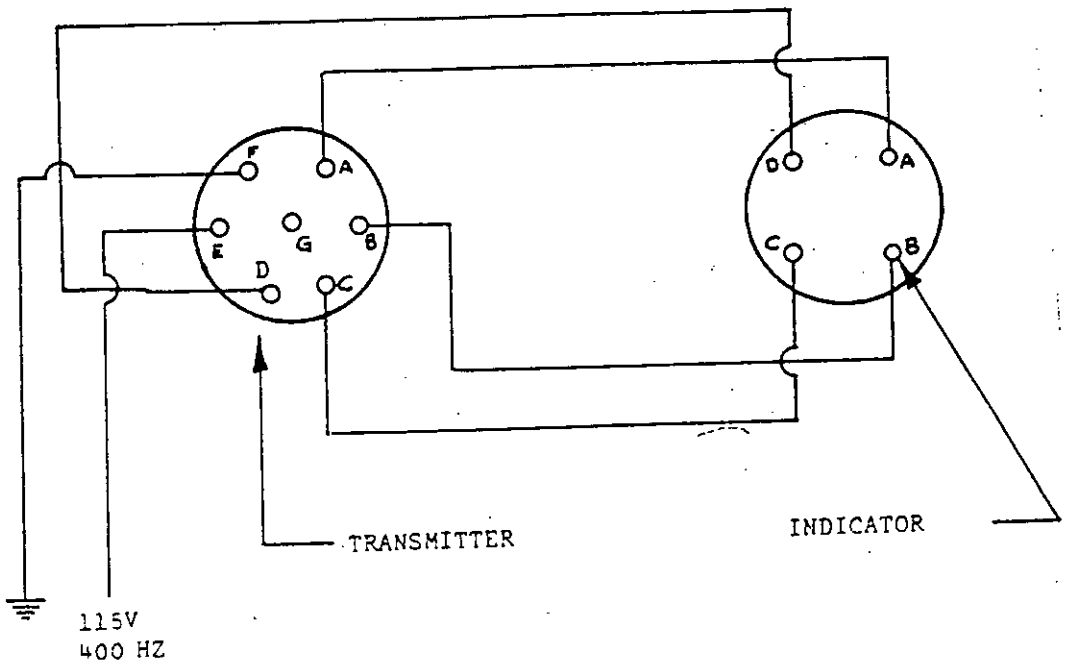


FIGURE 3. External wiring diagram.

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4.2 General. Sampling, inspection, and testing of the indicator shall be in accordance with MIL-I-7057, except as specified herein.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.4 First article inspection.

4.4.1 Test samples. The qualification test samples shall consist of three indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number.

4.4.2 Test report. The contractor shall prepare a test report and submit three copies to the qualifying activity for approval.

4.4.3 First article inspection. The first article inspection tests shall consist of all the tests of this specification, including applicable tests of MIL-I-7057. Unless otherwise specified by the qualifying activity, tests shall be performed in the order listed.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of:

a. Individual inspection
b. Sampling plan inspection.

4.5.1 Individual inspection. The test of 4.6.1 shall be performed in place of the scale error and friction test of MIL-I-7057.

4.5.2 Sampling plans and tests.

4.5.2.1 Sampling plan A. In addition to the sampling plan A tests of MIL-I-7057, the thermal shock tests of 4.6.6 and the counter cycling tests of 4.6.7 shall be performed.

4.5.2.2 Sampling plan B. The thermal shock and high altitude-low temperature tests of MIL-I-7057 shall not be required.

4.6 Test methods.

4.6.1 Scale error and friction. During the scale error test, the indicator can be connected to a master transmitter especially designed for this type of testing. The master transmitter shall not have an error of more than 0.05 circular degrees or shall be furnished with a correction table so that the correct transmitter position can be determined within 0.05 circular degrees. The corrected master transmitter readings required to provide the respective test indicator readings listed in Table I shall be set while the indicator is being gently tapped or vibrated. The indicator shall be considered out of tolerance when the difference between the test indicator reading and master synchro transmitter reading exceeds the tolerance shown in Table I. The test indicator friction error shall be measured by setting the indicator on the respective test points shown in Table I while the indicator is being gently tapped or vibrated. Then at each test point and without tapping or vibrating the indicator, the master synchro transmitter shall be advanced to an angular position which is just sufficient to cause the test indicator pointer to move. The test indicator dial shall also be scanned and observed for points of excessive

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friction at intermediate points. Friction error shall be considered to be the difference between the master synchro transmitter setting required to produce the test indicator setting while under vibration or tapping and the master synchro transmitter setting required to move the pointer without vibration or tapping.

friction at intermediate points. Friction error shall be considered to be the difference between the master synchro transmitter setting required to produce the test indicator setting while under vibration or tapping and the master synchro transmitter setting required to move the pointer without vibration or tapping.

4.6.2 Position error. The tolerance specified in Table II shall apply.

4.6.3 Vibration error. The tolerance specified in Table II shall apply. Upon completion of the vibration test, the test specified under 4.6.1 shall be repeated.

TABLE I. Indicator pressure ratio tolerances.

PRESSURE RATIO	TOLERANCE ± PRESSURE RATIO		
	ROOM TEMP	LOW TEMP	HIGH TEMP
1.2	0.007	0.007	0.007
1.5			
2.0	0.007	0.007	0.007
2.5			
3.0	0.007	0.007	0.007
3.4			

TABLE II. Indicator test tolerances.

TEST	TOLERANCE ± PRESSURE RATIO
Frequency and Voltage Variation	0.005
Position Error	0.005
Friction:	
Room Temperature	0.010
Low Temperature	0.015
High Temperature	0.010
Vibration:	
Pointer Oscillation	0.010
Pointer Variation	0.010
Endurance	0.010

4.6.4 Low temperature. While at the test temperature, the indicator shall be tested for scale error and friction. Errors shall not exceed the respective tolerances specified in Tables I and II. The indicator shall not require more than five seconds to come within the specified tolerance.

4.6.5 High temperature. While at the test temperature, the indicator shall be tested for scale error and friction. Errors shall not exceed the respective tolerances specified in Tables I and II.

4.6.6 Thermal shock. The indicator shall be subjected to six cycles of immersion in water at a temperature of $85^{o} \pm 2^{o}$ C and then in water of $5^{o} \pm 2^{o}$ C without evidence of moisture penetration or damage to the coating or case. The length of time for each immersion shall be 30 minutes, with no more than five seconds between immersions. After six continuous

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cycles have been completed, the indicator shall be subjected to the sealing test of MIL-I-7057. Leakage in excess of that permitted or fogging of the inner side of the cover glass shall be cause for rejection.

4.6.7 Counter cycling. The control knob shall be pushed in and turned continuously in order to move the takeoff index and counter from 1.3 ± 0.1 to 3.3 ± 0.1 and return to 1.3 ± 0.1 pressure ratio units. The control knob shall then be pulled out and turned continuously in order to move the cruise index and counter 1.3 ± 0.1 to 3.3 ± 0.1 and return to 1.3 ± 0.1 pressure ratio units. This procedure shall be considered one cycle of operation. The total cycling period shall be a minimum of 2,500 cycles and shall be conducted at a cycling rate of 1 ± 0.2 cpm. The last 500 cycles shall be conducted over the entire range of the scale and shall cause the slip clutch to engage for at least one complete revolution of the set knob at both ends of the scale. If stops are used instead of slip clutches, the index shall be driven against the stop with a minimum torque of 24 ounce inches. Prior to the cycling test, the torque required to turn the control knob shall be measured and recorded. The allowable torque required to turn the control knob throughout the takeoff and cruise range of 1.2 to 3.4 pressure ratio shall not exceed 6 ± 3 ounce inches. Upon completion of the 2,500 cycle testing, physical and functional operation inspection shall be conducted to assure that binding, deformation, and malfunction is not evident. The torque required to turn the control knob throughout the takeoff and cruise range of 1.2 to 3.4 pressure ratio shall not exceed the measurements recorded at the start of the cycling test by more than two ounce inches.

4.6.8 Low temperature exposure. While at the test temperature, the scale error and friction shall not exceed the tolerances specified in Tables I and II. After the indicator has been maintained at room temperature for not less than four hours, scale error and friction shall not exceed the tolerances specified in Tables I and II.

4.6.9 High temperature exposure. While at the test temperature, the scale error and friction shall not exceed the tolerances specified in Tables I and II. After the indicator has been maintained at room temperature for not less than four hours, scale error and friction shall not exceed the tolerances specified in Tables I and II.

4.6.10 Fogging. The indicator shall have been energized and in a controlled ambient of $70^{\circ} \pm 2^{\circ}$ C for a minimum of one hour. At the end of this period and while at this temperature, an ice cube made of distilled water and of nominal original size, one by one by one inch, with its flat surface one by one inch, shall be rubbed over a major area of the indicator glass face from one to two minutes. If the ice cube melts to an unusable size before this test is completed, continue by using a second ice cube. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of internal water or oil fog. Evidence of fogging shall be cause for rejection.

4.6.11 Humidity. Humidity tests shall be as specified in MIL-I-7057.

4.6.12 Electrical zero. Connections between the indicator and the electrical zero transformer shall conform to Figure

4.7 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see section 6.2).

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

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5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in Quantity Unit Pack (QUP) of one each. Each item will be provided a preservation method IC-1 in accordance with MIL-P-116.

5.1.2 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved in accordance with MIL-P-116 and MIL-STD-2073-1 to provide a method IC-1.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packing. Packing shall be level "A," "B," or "C" as specified (see 6.2).

5.2.1 Container-Cushioning. Unit container requirements shall conform to MIL-B-121, barrier material, unless otherwise specified by the contracting activity. Unit container shall be specified by the contracting activity. Unit container shall be large enough to allow for the application of cushioning/wrap material, of sufficient density and thickness, to protect the unit container from punctures. Intermediate/shipping containers shall be large enough to allow for the application of sufficient cushioning material, between the unit container and intermediate/shipping container. Cushioning material shall be of sufficient density and thickness to ensure shock transmission does not exceed peak values in "G's" established for the item.

5.2.2 Level A. Fiberboard containers do not meet level "A" container criteria.

5.2.3 Level B. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-1672, weather-resistant unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size, and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedure and contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked in accordance with special markings required by the contracting activity and MIL-STD-129.

5.4 Inspection and test. Test of methods of preservation shall be accomplished in accordance with section 4 of MIL-P-116 to ensure compliance with section 5 of this specification. Packaging tests shall be conducted in accordance with rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The indicator covered by this specification is intended for use to indicate the rate of, and to initiate signal for the summation of, flow of fuel in pounds per hour consumed by an aircraft turbo-jet engine.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODIIS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see section 2.1)

c. Levels of preservation, packaging, and packing required (see section 5)

d. Number of first article samples (see 4.4).

6.3 First article. When the first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3.

6.4 Subject term (key words) listing. Recommended key words are:

Flow
Fuel
Turbo-jet

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 71

Project number: 6620-F550

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